

Population pharmacokinetic drug–drug interaction pooled analysis of existing data for rifabutin and HIV PIs

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Objectives: Extensive but fragmented data from existing studies were used to describe the drug–drug interaction between rifabutin and HIV PIs and predict doses achieving recommended therapeutic exposure for rifabutin in patients with HIV-associated TB, with concurrently administered PIs.

Methods: Individual-level data from 13 published studies were pooled and a population analysis approach was used to develop a pharmacokinetic model for rifabutin, its main active metabolite 25-O-desacetyl rifabutin (des-rifabutin) and drug–drug interaction with PIs in healthy volunteers and patients who had HIV and TB (TB/HIV).

Results: Key parameters of rifabutin affected by drug–drug interaction in TB/HIV were clearance to routes other than des-rifabutin (reduced by 76%–100%), formation of the metabolite (increased by 224% in patients), volume of distribution (increased by 606%) and distribution to the peripheral compartment (reduced by 47%). For des-rifabutin, clearance was reduced by 35%–76% and volume of distribution increased by 67%–240% in TB/HIV. These changes resulted in overall increased exposure to rifabutin in TB/HIV patients by 210% because of the effects of PIs and 280% with ritonavir-boosted PIs.

Conclusions: Given together with non-boosted or ritonavir-boosted PIs, rifabutin at 150 mg once daily results in similar or higher exposure compared with rifabutin at 300 mg once daily without concomitant PIs and may achieve peak concentrations within an acceptable therapeutic range. Although 300 mg of rifabutin every 3 days with boosted PI achieves an average equivalent exposure, intermittent doses of rifamycins are not supported by current guidelines.

Introduction

TB and HIV are major causes of mortality worldwide. In HIV-associated TB, concurrent ART and TB treatment is associated with substantially reduced mortality compared with TB treatment alone.¹ However, for patients requiring HIV PIs, combined treatment for HIV and TB is complicated by drug–drug interactions with the anti-TB rifamycins. Anti-TB regimens

are better when they include a rifamycin, especially in HIV-infected patients,² and rifampicin is established in TB treatment programmes worldwide. However, rifampicin potentially induces the metabolism of PIs, necessitating dose adjustments, and there are concerns about the safety and efficacy of the currently used approaches.^{3–10} The potential for drug interactions is high in HIV-infected patients who are treated with multiple drugs.

Rifabutin, an alternative to rifampicin, has the important advantage of not substantially affecting the concentrations of PIs.¹¹ Although rifabutin may be an effective anti-TB agent in patients treated concurrently with PI-based ART, the optimal dose and dose interval for rifabutin are unknown. Based on limited evidence,^{12–15} a 50%–75% dose reduction has been recommended when rifabutin is given concurrently with ritonavir-boosted PIs.³ Furthermore, rifabutin concentrations may be lower in patients than healthy volunteers.^{14,16} However, rifabutin has dose-related toxicity.^{17,18} Serious toxicity, including uveitis, polyarthralgia, polymyalgia and neutropenia, may occur in patients treated with high doses, especially in association with enzyme inhibitors that may cause disproportionate increases in the concentrations of rifabutin metabolites such as 25-*O*-desacetyl rifabutin (des-rifabutin).^{11,19,20}

Des-rifabutin predominates among several metabolites of rifabutin, is formed by arylacetamide deacetylase and has activity similar to the parent compound.²¹ In healthy volunteers, the plasma des-rifabutin:rifabutin AUC ratio is 0.1 in the absence of inhibitors.^{22,23} Rifabutin is extensively distributed to the tissues because it has a wide volume of distribution and long half-life (>30 h) in HIV-infected patients.²⁴

Rifabutin also has high between-subject variability in its pharmacokinetics (coefficient of variation for clearance, 32%).²⁵ Therefore, the impact of dose interval and inhibition of its metabolism by PIs on the steady-state pharmacokinetics of the drug in plasma and tissues is of relevance to the understanding of pharmacokinetic–pharmacodynamic relations.

A population pharmacokinetic analysis approach can be used to analyse retrospectively collected data in a pooled analysis (meta-analysis)^{26–28} and relations between demographics and parameters can be identified. The purpose of the present study was to characterize the pharmacokinetics of rifabutin and its main metabolite, des-rifabutin, in the presence of several different PIs, using a population pooled analysis approach to combine extensive but fragmented information from existing studies to model the pharmacokinetics of rifabutin concentrations in TB/HIV patients treated concurrently with PIs, and to determine the optimal dose and dose interval of rifabutin for prospective studies.

Methods

Data collection

In October 2009, a search was performed on PubMed and Google that identified 29 publications (abstracts and peer-reviewed papers)^{11,14,16–19,24,25,29–49} for a global analysis of rifabutin pharmacokinetics. In 14 publications (48%), the interactions with PIs were studied. To evaluate the difference in pharmacokinetics and interaction with PIs between TB and/or HIV patients and healthy volunteers, 8 publications were identified that established the pharmacokinetics of rifabutin in healthy volunteers^{11,19,32,35,37,38,44,49} and 16 (55%) publications were identified that reported the pharmacokinetics of rifabutin in TB patients with HIV (TB/HIV) or without HIV infection, or HIV patients without TB infection.^{14,16,24,25,30,31,33,34,36,37,39–43,50} Parent–metabolite relations between rifabutin and des-rifabutin were evaluated in six (20%) publications including healthy volunteers^{11,19,35,37,38,49} and nine (31%) publications including TB and/or HIV patients.^{30,33,34,36,40–43,47} The authors of the 29 identified publications were approached in 2009 and 2010 to contribute data to this pooled analysis. Pharmaceutical companies were also approached to contribute data from drug–drug interaction studies that

were not published in the public domain. Raw data from all individuals who participated in the existing studies were requested from the collaborating authors and companies. All studies had previously undergone ethics review and had been approved by institutional review boards. Information about the identity of participants in the previous studies was removed to make the data anonymous before data were transferred to our analysts (S. H., E. M. S. and R. N.). From these datasets, we extracted rifabutin and des-rifabutin concentrations, doses and times of drug administration, patient characteristics and information about coadministration with PIs. Observations were defined as either a rifabutin or des-rifabutin concentration measurement after a given dose and all observations within the same dosing interval were grouped as one occasion. Subsequent literature searches were performed until May 2015 for additional data sources, but data inclusion was stopped in 2011 to enable consistent model building.

Population pharmacokinetic pooled analysis

The population approach pooled analysis was based on collection of individual patient data from the different studies pooled into a single database. This enabled analysis of the combined data in a new population pharmacokinetic model.

A non-linear mixed-effect modelling approach was performed with software (NONMEM 7.2.0 to 7.3.0, ICON, Dublin, Ireland;⁵¹ Perl-speaks-NONMEM, PsN, Uppsala, Sweden, version 4.2.0 to 4.2.2, <http://psn.sourceforge.net>⁵²) to derive the population mean and variance for all pharmacokinetic parameters. Pharmacokinetic models were fitted to concentration–time data with standardized population pharmacokinetic methods.⁵³ NONMEM analyses were performed using the first-order method to increase speed of model development and the first-order conditional estimation method with interaction for the final stages. Nested models were hypothesis tested using the likelihood ratio test, in which the change in objective function value (OFV) approximated the χ^2 distribution ($\chi^2_{1,0.001} > 10.828$) and non-nested models were compared using the Akaike information criterion. Visual predictive checks with prediction and variability correction (pvcVPCs)⁵⁴ were used as a diagnostic tool throughout the analysis.

The model-building strategy was based on a previously published model that was developed from a smaller dataset.⁵⁵ The initiating model for the present analysis used two-compartment disposition for both rifabutin and des-rifabutin with first-order clearance parameterized in terms of apparent oral clearance of rifabutin by routes other than des-rifabutin (CL/F), apparent oral metabolism clearance of des-rifabutin (CL_m/F), apparent oral clearance of des-rifabutin (CL_m/FF_m), volume of distribution of the central compartment and the peripheral compartment for rifabutin (V_2 and V_3) and des-rifabutin (V_4 and V_5) and inter-compartmental clearance rate for rifabutin (Q) and des-rifabutin (Q_m). This model used first-order absorption parameterized as the rate constant of absorption (k_a) after a lag time and included a first-pass metabolism fraction (F_m) to des-rifabutin. The metabolism from parent to metabolite was included as apparent oral metabolism clearance of rifabutin into a metabolite compartment. The CL_m and V_4 were adjusted with the molecular weight to account for the difference in size between the two molecules.

Between-subject variability terms, with covariance, were tested on all parameters using exponential models. The covariance terms were evaluated on a linearized version of the model and included in the non-linear model when significant.⁵⁶ The relative bioavailability (F) varied for each individual between sampling occasions, given the same overall variance for each occasion.⁵⁷ The residual unexplained variability model was additive for logarithm-transformed concentration data, separately for rifabutin and des-rifabutin, corresponding to proportional error on untransformed data.

In the initial drug–drug interaction model, the interaction with PIs was modelled as present or absent, depending on whether PI coadministration occurred. The changes to the model with PI coadministration initially included a decrease in CL and CL_m , with both these interactions

strengthened when a ritonavir-boosted PI was given. Further parameter changes with PI coadministration were evaluated, allowing additional effects of ritonavir boosting and different effects for healthy volunteers and TB/HIV patients. Aside from PI coadministration, other covariates available for testing included body weight, age and sex for TB/HIV subjects versus healthy volunteers. The body weight was included on all disposition parameters *a priori* using allometric scaling.⁵⁸ All other covariate parameter relations were included as proportional on/off changes for categorical covariates or linear relations for continuous covariates. Parameter precision was obtained from a limited non-parametric bootstrap stratified by study and, for studies with several arms, stratified by arm.

Final estimated model parameters were used to calculate the expected average steady-state concentrations ($C_{av,ss}$) for rifabutin using equation (1) to identify dosing regimens that provided an exposure >4.5 mg·h/L (AUC_{0-24})^{30,34} or 0.187 mg/L ($C_{av,ss}$), which was previously associated with acquired rifamycin resistance. The parameter $C_{av,ss}$ was calculated using equation (1) and was transformed to an AUC for specific dosing intervals (τ) using equation (2), where CL_e is the systemic metabolism clearance to des-rifabutin:

$$C_{av,ss} = \frac{[F \cdot (1 - F_m) \cdot (\text{dose}/\tau)]}{(CL + CL_e)} \tag{1}$$

$$AUC_{\tau} = C_{av,ss} \cdot \tau \tag{2}$$

Steady-state peak concentrations ($C_{max,ss}$) for rifabutin and des-rifabutin were predicted from the final model using software (Berkeley Madonna, Version 8.3.18, Kagi, Berkeley, CA, USA).

Results

Data obtained

Nine collaborator teams contributed 13 datasets that were published previously.^{13,14,16,17,19,29,34,36,37,59} Furthermore, three datasets that were described previously in internal reports or conference proceedings were contributed from pharmaceutical companies.^{55,60-62}

The combined dataset used for the pharmacokinetic analysis of rifabutin and des-rifabutin, with and without coadministration of PIs, contained data from 251 subjects [mean age, 36.4 years; 188 male (75%); 164 HIV infected (65%); 141 HIV-infected

patients who had TB (56%)], totalling 7749 pharmacokinetic observations. The demographic details available varied between different studies, but all studies provided information about whether the subject was a healthy volunteer, had HIV infection or had TB (Table 1). There was no information available about other concomitantly given drugs in TB/HIV patients with known interaction, such as azoles and clarithromycin, except for PIs, and there were no toxicity data available.

Details of the design of the different studies that contributed data were available from the publications.^{13,14,16,19,29,34,36,37,55,59,61-63} Data points were not included in the combined dataset when details were missing about dose, time of observation, concentration, study arm or PI coadministration. Rifabutin and des-rifabutin pharmacokinetic observations were available after administration of rifabutin alone (without PIs) in 235 and 191 subjects, and varied numbers of patients contributed observations after administration of rifabutin with various PIs (Table 2). Fewer subjects contributed rifabutin and des-rifabutin observations with than without coadministration with a PI, due to higher frequency of dropout in the coadministration arm of several studies. A total of four (2%) patients, 51 (1.2%) rifabutin concentrations and 155 (4.3%) des-rifabutin concentrations were removed from the contributing studies because of incomplete information.

Rifabutin alone was administered: as a 150 mg dose once daily in 39 subjects and every 3 days in 1 subject; as a 300 mg dose once daily in 107 subjects, once off in 4 subjects, every second day in 10 subjects, once weekly in 11 subjects and every 3 days in 55 subjects; as a 450 mg dose every 3 days in 1 subject; and as a 600 mg dose every 3 days in 7 subjects. Rifabutin was administered as a 150 mg dose once daily in 42 subjects and every 3 days in 7 subjects together with ritonavir, darunavir/ritonavir, saquinavir/ritonavir, indinavir or lopinavir/ritonavir; as a 300 mg dose once daily in 10 subjects, once weekly in 10 subjects, every 2 days in 8 subjects and every 3 days in 27 subjects together with saquinavir/ritonavir, indinavir, lopinavir/ritonavir, nelfinavir or amprenavir/ritonavir; and as a 600 mg dose every 3 days in 1 subject together with indinavir and in 1 subject together with nelfinavir. Observations were made in patients mainly (89%) on three occasions.

Table 1. Demographic characteristics of the study subjects who had rifabutin and des-rifabutin concentrations measured (total number of subjects in the dataset was 251)

Characteristic	Number of subjects (%) out of 251 subjects who had missing information on that characteristic	Number of subjects (%) out of 251 subjects in dataset with this characteristic or mean ± SD (range)
TB	0 (0)	141 (56.2)
HIV	0 (0)	164 (65.3)
Healthy volunteer	0 (0)	87 (34.7)
TB/HIV		141 (56.2)
Sex (female/male)	15 (6.0)	48 (19.1)/188 (74.9)
Race (white/black/other/Hispanic)	76 (30.3)	57 (22.7)/75 (30.0)/4 (1.5)/39 (15.5)
Weight (kg)	33 (13.1)	68 ± 14 (40.8–119.7)
Age (years)	14 (5.6)	36 ± 9 (18.0–67.0)
Height (cm)	52 (20.7)	170 ± 11 (135.0–198.0)
CD4+ count in HIV patients (cells/μL)	139 (55.4)	217 ± 51 (11.0–928.0)

Table 2. Patients and rifabutin and des-rifabutin pharmacokinetic measurements contributing to the pooled analysis

Pharmacokinetic observation	Rifabutin, n (%)	Des-rifabutin, n (%)
Number of patients	251	235 (93.6)
Patients with data when rifabutin was not coadministered with PI	235 (93.6)	191 (76.1)
Patients with data when rifabutin was coadministered with PI	136 (54.1)	122 (48.6)
yes	79 (31.4)	75 (29.8)
no	57 (22.7)	47 (18.7)
PI used		
ritonavir	5 (1.9)	5 (1.9)
darunavir/ritonavir	13 (5.1)	13 (5.1)
saquinavir/ritonavir	19 (7.5)	18 (7.2)
indinavir	22 (8.7)	22 (8.7)
lopinavir/ritonavir	41 (16.3)	39 (15.5)
nelfinavir	35 (13.9)	25 (9.9)
amprenavir/ritonavir	1 (0.4)	0 (0)
Total number of observations	4168	3581 (85.9)
Observations when rifabutin was not coadministered with PI	2046 (49.1)	1682 (40.4)
Observations when rifabutin was coadministered with PI	2122 (50.9)	1899 (45.5)
yes	1365 (32.7)	1299 (31.2)
no	757 (18.1)	600 (14.4)
PI used		
ritonavir	60 (1.4)	60 (1.4)
darunavir/ritonavir	178 (4.2)	178 (4.2)
saquinavir/ritonavir	562 (13.5)	533 (12.7)
indinavir	465 (11.2)	465 (11.2)
lopinavir/ritonavir	558 (13.4)	528 (12.6)
nelfinavir	292 (7.0)	135 (3.2)
amprenavir/ritonavir	7 (0.2)	0 (0)

Population pharmacokinetic pooled analysis

Structural and variance

For structural additions to the initial model (Figure 1), the main extension was the inclusion of a transit compartment model described by number of transit compartments (N) and mean transit time instead of a lag-time parameter to describe the absorption of rifabutin (OFV change, -637 points; 1 degree of freedom). The variance structure was changed from a complex stochastic model with different between-subject variability for different subpopulations (e.g. HIV patient data were more variable than data from healthy volunteers) to a full correlation matrix and between-subject variability on all structural model parameters, which resulted in a significantly improved model (OFV change, -918 ; 65 degrees of freedom). Between-subject variability was not included for parameters describing parameter–covariate relations. The pharmacokinetics of rifabutin were highly variable, with between-subject variability on parameters usually $>50\%$.

Covariate and drug–drug interaction

The effect of coadministration with a PI was retained from the initial model on CL and CL_m and extended by differentiating this effect for ritonavir-boosted and non-boosted PIs. Age and sex had no significant effect on any parameters during the first linearized stepwise covariate model (lin-SCM) building step (detailed methods available as Supplementary data at JAC Online). In the second lin-SCM step, the following covariates were tested on all structural parameters: coadministration with a PI, ritonavir boosting and TB/HIV versus healthy volunteers, with some of these effects specific for TB/HIV patients compared with healthy volunteers, and covariate–parameter relations were included in the final model (Figure 1). The final parameter estimates and full variance–covariance matrix from the final model were determined (Tables S1 and S2, available as Supplementary data at JAC Online). To account for the complexity of the model and facilitate the interpretation of the parameters, estimates were translated into parameter values for healthy volunteers and TB/HIV patients when rifabutin was administered alone, together with a non-boosted PI or a ritonavir-boosted PI (Table 3). The model described pharmacokinetic results comparable to results of previous studies^{24,64} and predicted a terminal half-life of 34 h for rifabutin in healthy volunteers and 29 h in TB/HIV patients. The effects of the drug–drug interaction with PIs on rifabutin and des-rifabutin pharmacokinetic parameters were converted into percentage reductions or increases in the parameter value compared with the parameter values without PI coadministration (Table S3). Pharmacokinetic parameters for rifabutin and des-rifabutin were similar between healthy volunteers and TB/HIV patients when rifabutin was taken alone, except for a smaller volume of distribution for the patient population. When rifabutin was coadministered with a PI, most parameters were changed by $>20\%$ (comparison of parameters in Tables 3 and S3). Most importantly, rifabutin coadministration with a PI caused a CL decrease of $>76\%$, a V_2 increase of 6-fold, a Q decrease of 47% and an increase in transformation to des-rifabutin of 50% in healthy volunteers and 224% in patients. The increase in apparent oral metabolism clearance to des-rifabutin could be caused by a reduced sequential metabolism in the presence of PIs. Des-rifabutin had: a decrease in CL_m of 35% when rifabutin was coadministered with a non-boosted PI and $>76\%$ when rifabutin was given with a ritonavir-boosted PI; an increase in V_4 of $>67\%$ for patients; and decreases in Q_m and F_m .

Model evaluation

The pvcVPC for the final model was stratified to present data for healthy volunteers and patients separately (Figure 2). The pvcVPC showed that the model adequately described the average exposure over time for both rifabutin and des-rifabutin. The description of rifabutin and des-rifabutin pharmacokinetics and variability in TB/HIV patients was adequate. A more detailed pvcVPC stratified for coadministration with non-boosted and ritonavir-boosted PIs showed that the model adequately described the average exposure over time of all subgroups; patient data were better characterized by the model than healthy volunteer data, in which model-predicted variability exceeded observed variability (Figure S1).

Estimates of parameter precision for the final model were obtained using a limited bootstrap method. The analysis was limited to 13 bootstrap samples owing to long runtimes

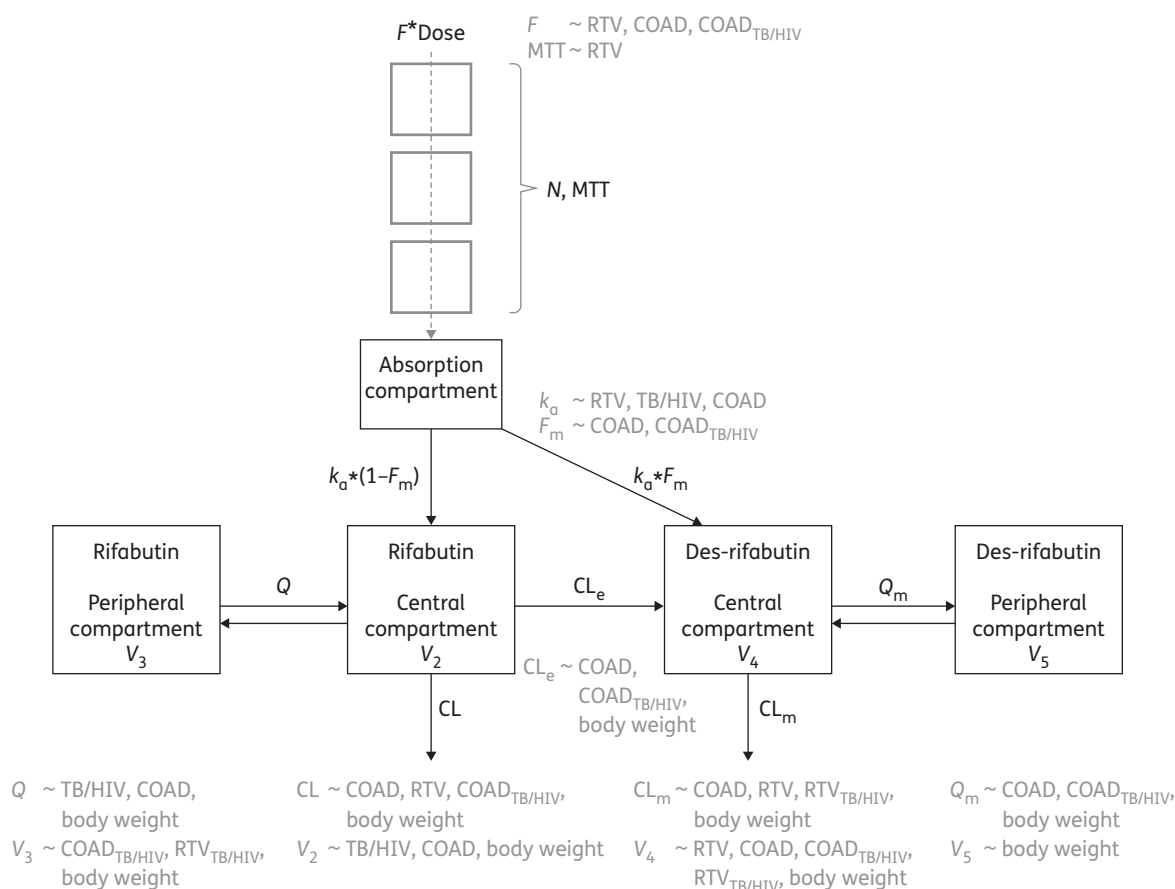


Figure 1. Schematic of the final pharmacokinetic model for rifabutin, des-rifabutin and PIs, including all parameters and covariates associated with specific parameters. Parameters affected by drug interaction with PIs are shown in grey. Covariate relations were included before performing lin-SCM. All disposition parameters were allometrically scaled with the factor $F_{\text{CLbody weight}} = (\text{body weight}/70)^{0.75}$ on all clearance parameters and $F_{\text{Vbody weight}} = \text{body weight}/70$ on all volume parameters. CL, systemic clearance of rifabutin to routes other than to des-rifabutin; CL_e , systemic metabolism to des-rifabutin; CL_m , systemic clearance of des-rifabutin; COAD, coadministration with a PI; $\text{COAD}_{\text{TB/HIV}}$, coadministration with a PI in TB/HIV patients; F , bioavailability; F_m , fraction first-pass metabolism; TB/HIV, patients with HIV and TB; k_a , absorption rate constant; MTT, mean transit time; N , number of transit compartments; Q , intercompartmental clearance rate for rifabutin; Q_m , intercompartmental clearance rate for des-rifabutin; RTV, coadministration with ritonavir or ritonavir-boosted PI; $\text{RTV}_{\text{TB/HIV}}$, coadministration with ritonavir or ritonavir-boosted PI in TB/HIV patients; V_2 , volume of distribution, central compartment, rifabutin; V_3 , volume of distribution, peripheral compartment, rifabutin; V_4 , volume of distribution, central compartment, des-rifabutin; V_5 , volume of distribution, peripheral compartment, des-rifabutin.

(one estimation parallelized on 32 cores obtained stable OFV in 10 days). The relative standard errors for most parameters were $<20\%$, except for some covariate-parameter relation parameters, such as: V_2 and HIV; k_a and coadministration with PIs and HIV; F and coadministration with PIs; F and coadministration with ritonavir-boosted PIs; CL_e and coadministration with PIs; F_m and coadministration with PIs in HIV; and V_4 and coadministration with PIs (Table S1).

Dose recommendations

The final model parameters were used to calculate the expected average steady-state concentration [$C_{\text{av,ss}}$ (mg/L)] for rifabutin (Table 4) and model-predicted steady-state peak concentrations ($C_{\text{max,ss}}$) for rifabutin and des-rifabutin (Table 5) for a selection of dosing regimens when administered alone or together with PIs. These parameters were compared with reported pharmacokinetic

values for rifabutin and the applied dosing schedules (Table S4).^{11-16,19,20,25,31,33-38,49,55,64-69} The expected $C_{\text{av,ss}}$ after rifabutin (300 mg once daily) alone determined with the model was 0.17 mg/L for a typical patient or healthy volunteer, corresponding to an AUC_{0-24} of 4.08 mg·h/L. The rifabutin dosing regimen that matched the exposure of a 300 mg daily dose without PI was 150 mg once daily for patients (300 mg every 3 days for healthy volunteers) when coadministered with a non-boosted PI and 300 mg every 3 days for patients (150 mg every 2 days for healthy volunteers) when given with a ritonavir-boosted PI. A target $C_{\text{av,ss}} \geq 0.187$ mg/L ($\text{AUC}_{\text{ss},0-24}$, 4.5 mg·h/L)^{30,34} for the typical patient was best achieved using a dosage of 150 mg once daily when given together with a non-boosted PI or alternating doses of 150 and 300 mg every second day when given together with a ritonavir-boosted PI.

Model-predicted steady-state peak concentrations ($C_{\text{max,ss}}$) for rifabutin and des-rifabutin after 300 mg of rifabutin once

Table 3. Pharmacokinetic parameter estimates for rifabutin and des-rifabutin in the final model with and without drug–drug interaction with PIs

Parameter	Units	Rifabutin alone		Rifabutin with non-boosted PI		Rifabutin with ritonavir-boosted PI	
		healthy volunteers	TB/HIV patients	healthy volunteers	TB/HIV patients	healthy volunteers	TB/HIV patients
Rifabutin							
apparent oral clearance to routes other than des-rifabutin, CL/F	L/h/70 kg	58.80	58.80	13.88	0.00	12.00	0.00
apparent oral metabolism clearance to des-rifabutin, CL _e /F	L/h/70 kg	5.76	5.76	8.65	18.68	8.65	18.68
central volume of distribution, V ₂ /F	L/70 kg	6.55	4.59	46.30	32.44	46.30	32.44
absorption rate constant, k _a	/h	0.25	0.23	0.28	0.26	0.14	0.13
mean transit time, MTT	h	2.16	2.16	2.16	2.16	1.49	1.49
number of transit compartments, N		7.15	7.15	7.15	7.15	7.15	7.15
bioavailability, F		0.97	0.97	1.03	0.60	1.38	0.80
peripheral volume of distribution, V ₃ /F	L/70 kg	1580	1580	1580	571.9	1580	1395.6
inter-compartmental clearance, Q/F	L/h/70 kg	62.61	90.08	33.24	47.82	33.24	47.82
Des-rifabutin							
apparent oral clearance, CL _m /F/F _m	L/h/70 kg	122.00	122.00	79.06	79.06	8.78	28.81
central volume of distribution, V ₄ /F/F _m	L/70 kg	37.30	37.30	91.01	126.78	31.58	62.56
inter-compartmental clearance, Q _m /F/F _m	L/h/70 kg	71.80	71.80	25.06	74.17	25.06	74.17
peripheral volume of distribution, V ₅ /F/F _m	L/70 kg	1220	1220	1220	1220	1220	1220
first-pass metabolism fraction, F _m		0.09	0.09	0.05	0.10	0.05	0.10

CL, systemic clearance of rifabutin to routes other than des-rifabutin; CL_e , systemic metabolism to des-rifabutin; CL_m , systemic clearance of des-rifabutin; F , bioavailability; F_m , fraction first-pass metabolism; k_a , absorption rate constant; MTT, mean transit time; N , number of transit compartments; Q , inter-compartmental clearance rate for rifabutin; Q_m , inter-compartmental clearance rate for des-rifabutin; V_2 , volume of distribution, central compartment, rifabutin; V_3 , volume of distribution, peripheral compartment, rifabutin; V_4 , volume of distribution, central compartment, des-rifabutin; V_5 , volume of distribution, peripheral compartment, des-rifabutin.

All disposition parameters were allometrically scaled with the factor $F_{CL\text{body weight}} = (\text{body weight}/70)^{0.75}$ on all clearance parameters and $F_{V\text{body weight}} = \text{body weight}/70$ on all volume parameters.

daily alone were 0.43 and 0.37 mg/L for the typical healthy volunteer and patient (Table 5), consistent with results of published reports (Table S4).⁷⁰ To reach a similar $C_{\text{max_ss}}$ when administered with a non-boosted PI or a ritonavir-boosted PI, a 50% reduction in the daily dose of rifabutin to 150 mg once daily would be required. The time to achieve the $C_{\text{max_ss}}$ was 3.5-fold longer with ritonavir-boosted PI coadministration than with non-boosted PIs. Substantial accumulation of des-rifabutin was noted, with the rifabutin $C_{\text{max_ss}}$ to des-rifabutin $C_{\text{max_ss}}$ ratio decreasing from 10:1 with rifabutin alone to 5:1 when coadministered with a non-boosted PI and 2:1 when coadministered with a ritonavir-boosted PI. A combined rifabutin and des-rifabutin $C_{\text{max_ss}} > 1$ mg/L,⁴⁰ the suggested threshold for greater risk of adverse drug reactions, would be uncommon in TB/HIV patients unless the dosage was 300 mg once daily together with ritonavir-boosted PIs (Table 5).

Discussion

The results showed a successful pharmacokinetic model for rifabutin and des-rifabutin including covariates together with a drug–drug interaction model for coadministration with PIs, using the extensive but fragmented data from existing studies. The model-based population pooled analysis successfully described the pharmacokinetics and the variability of rifabutin and des-rifabutin exposure in patients with HIV-associated TB and healthy

volunteers. The pharmacokinetics of rifabutin and des-rifabutin were found to be highly variable between subjects and body weight was the only patient characteristic that explained some of the variability. The pharmacokinetic parameters were more variable in patients than in healthy volunteers (Figure S1). Wide pharmacokinetic variability has been reported previously (Table S4). The reported pharmacokinetic values varied between studies. Reports of differences in exposure between healthy volunteers and TB/HIV patients had caused uncertainty in optimal drug dosing in patients who required concomitant anti-TB and HIV treatment. Here we have presented results for both healthy volunteers and TB/HIV-coinfected patients (Tables 3–5 and S3), which should support decisions for future clinical trials.

This pooled analysis showed that the average exposure was similar in patients and healthy volunteers when rifabutin was given alone. When PIs or ritonavir-boosted PIs were coadministered with rifabutin, the exposure of patients to rifabutin was decreased by 65% and 60% compared with the exposure of healthy volunteers (Table 4). Literature review showed no previous study that evaluated the pharmacokinetics of rifabutin simultaneously in healthy volunteers and TB/HIV patients, but this pooled analysis of data from both groups enabled estimation of the differences.

The present model predicted that a decrease of 50% or 67% in the rifabutin dose is required when rifabutin is coadministered

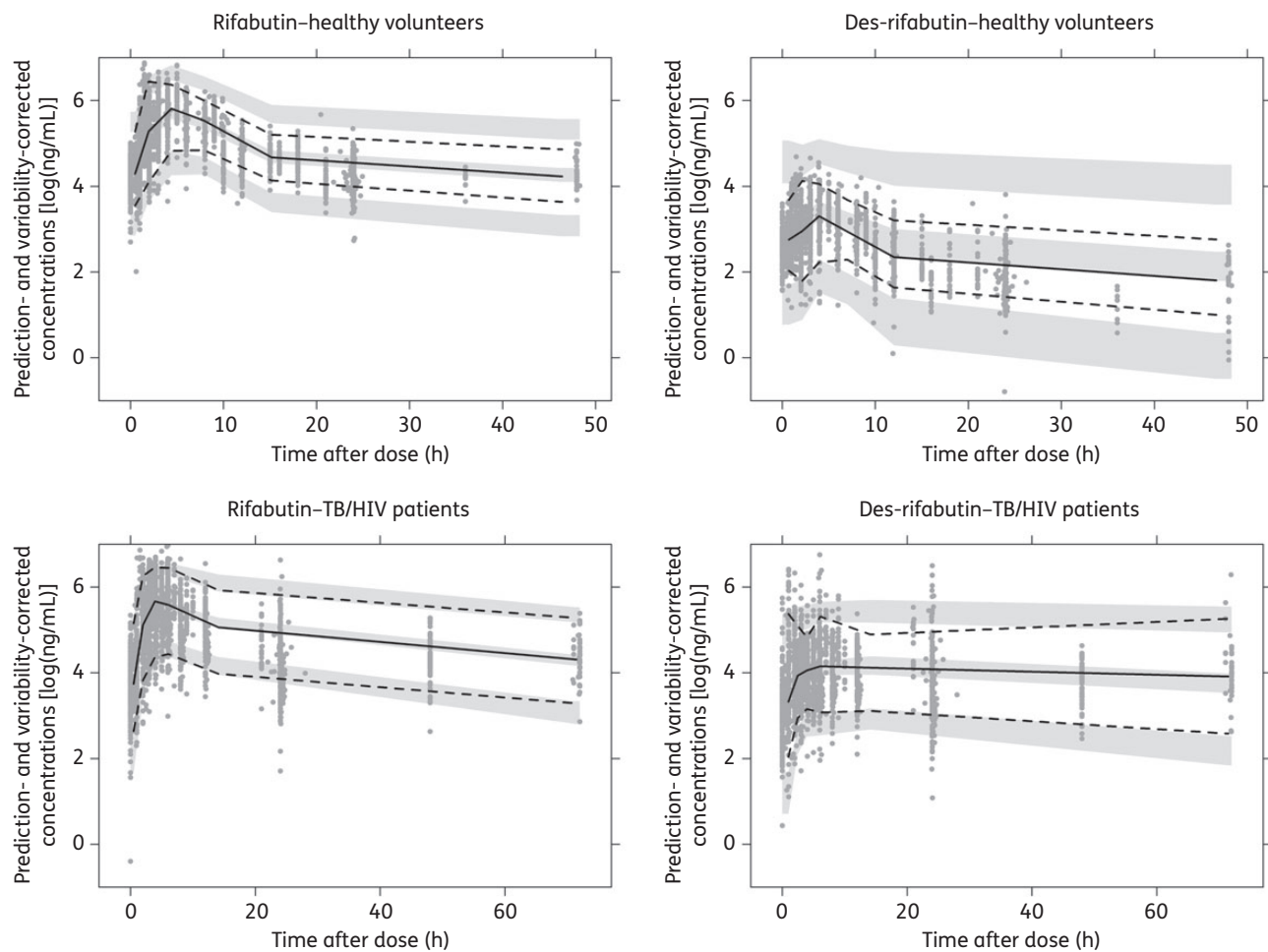


Figure 2. pvcVPC for rifabutin (left-hand panels) and des-rifabutin (right-hand panels) for the final model. The pvcVPC was stratified to present data separately for healthy volunteers (top panels) and TB/HIV patients (bottom panels). Raw data are represented as grey dots. Top, middle and bottom lines represent 95th, 50th and 5th percentiles of the observations. Grey shaded areas are 90% CIs for the 95th, 50th and 5th percentiles of the simulated data.

Table 4. Relation between rifabutin dosing regimen, use of PIs and expected average steady-state rifabutin concentration

Rifabutin dosing regimen	Rifabutin alone		Rifabutin with non-boosted PI		Rifabutin with ritonavir-boosted PI	
	healthy volunteers	TB/HIV patients	healthy volunteers	TB/HIV patients	healthy volunteers	TB/HIV patients
300 mg once daily	0.17	0.17	0.55	0.36	0.80	0.48
150 mg once daily	0.08	0.08	0.27	0.18	0.40	0.24
150 mg every 2 days	0.04	0.04	0.14	0.09	0.20	0.12
300 mg every 3 days	0.06	0.06	0.18	0.12	0.27	0.16
Alternating 150 and 300 mg every 2 days	0.06	0.06	0.20	0.13	0.30	0.18

Data are reported as the average steady-state rifabutin concentration (C_{av_ss} , mg/L).

with a non-boosted or a ritonavir-boosted PI, consistent with previous studies. The dosing regimens suggested for coadministration with a PI matched the exposure after 300 mg of rifabutin alone once daily. A typical patient receiving 300 mg of rifabutin daily without PIs would have exposure slightly below the suggested lower limit ($AUC_{0-24} > 4.5$ mg/L).^{30,34} However, half of

Table 5. Relation between rifabutin dosing regimen, use of PIs and expected steady-state rifabutin and des-rifabutin peak concentration

Rifabutin dosing regimen	Rifabutin alone				Rifabutin with non-boosted PI				Rifabutin with ritonavir-boosted PI			
	rifabutin		des-rifabutin		rifabutin		des-rifabutin		rifabutin		des-rifabutin	
	healthy volunteers	TB/HIV patients	healthy volunteers	TB/HIV patients	healthy volunteers	TB/HIV patients	healthy volunteers	TB/HIV patients	healthy volunteers	TB/HIV patients	healthy volunteers	TB/HIV patients
	0.43	0.37	0.04	0.04	1.16	0.64	0.13	0.13	1.26	0.69	0.97	0.39
300 mg once daily	0.22	0.18	0.02	0.02	0.58	0.32	0.06	0.07	0.63	0.35	0.48	0.20
150 mg once daily	0.19	0.16	0.02	0.02	0.50	0.25	0.05	0.05	0.49	0.25	0.29	0.11
150 mg every 2 days	0.38	0.30	0.04	0.03	0.94	0.47	0.10	0.08	0.89	0.44	0.46	0.17
300 mg every 3 days												

Data are reported as the average steady-state rifabutin or des-rifabutin peak concentration (C_{max_ss} , mg/L).

the patients may have rifabutin exposure below the exposure required to avoid acquired rifamycin resistance. Similarly, both suggested doses, when given together with a PI, would provide exposure slightly below the target minimum exposure. To recommend alternative dosages, new formulations and strengths of rifabutin need to be made available. Other studies have reported average exposures below $AUC_{0-24} > 4.5$ mg/L after 300 mg of rifabutin once daily in patients^{13,15,16,31,37} (Table S4). The predicted C_{max_ss} for the average TB/HIV patient after the typical adult dose of rifabutin (300 mg once daily) is slightly below the proposed lower limited of 0.45 mg/L.^{30,70} The same was predicted for a dosage of 150 mg of rifabutin once daily together with non-boosted or ritonavir-boosted PIs. Achieving peak concentrations within the range 0.45–0.9 mg/L in more than half of the patients would require new formulations, such as 100 or 200 mg capsules, and additional studies would be required to evaluate new dosages or formulations.

There is clinical concern about intermittent dosing for rifabutin because of the risk of therapeutic failure caused by selection of resistance in patients who have low CD4 counts and a disseminated bacillary burden, such as HIV-positive patients, which is markedly higher than in HIV-negative patients with pulmonary TB. Therefore, rifabutin exposure requirements may not be the same in these two patient groups: a higher number of bacilli may be associated with a higher probability of selection of naturally resistant mutants. Furthermore, HIV patients frequently present with low compliance owing to the complexity of their treatment. Dosing complexity increases with intermittent rifabutin dosing and risk of resistance may increase therapeutic failure in sicker patients compared with patients who have more immune system support or patients during maintenance treatment. Intermittent dosing may be more adequate in HIV-negative patients to avoid adverse events. Dosing of rifabutin may be adapted to the disease status of the patients.

The clinical concern about intermittent dosing of rifabutin is matched by current guidelines from the US CDC,³ which recommends that 150 mg of rifabutin once daily should be given with a ritonavir-boosted PI in adults. However, there are limited safety data with this dosage and combination and it is unknown whether the increase in concentration of rifabutin and des-rifabutin that may result from this dosage may cause increased risk of developing uveitis, neutropenia or hepatotoxicity; patients taking this combination should be monitored for rifabutin-related toxicities.¹⁹ Limited data are available about the exposure–toxicity relationship, suggesting that the dosage should be decreased when the peak concentration (or the combined concentration of rifabutin and des-rifabutin) is > 1 mg/L.^{40,70} The present model predicts that this potentially toxic peak concentration may occur for patients who receive 300 mg of rifabutin once daily together with a ritonavir-boosted PI (Table 5) and may be problematic for healthy volunteers in future clinical trials under dosing regimens in which several PIs are coadministered. The upper boundary has not been confirmed and serious toxicity may be common with rifabutin dosages of 150 mg from 3 times/week to daily.^{13,71} When considering the suggested dosages, which aim to match the exposure after a 300 mg once-daily rifabutin dose alone and the minimum exposure target proposed previously,^{30,34} 50% of patients will achieve an exposure above these targets. Rifabutin exposure increases with coadministration of PIs and a disproportionate increase in des-rifabutin exposure also occurs.

The increased des-rifabutin exposure may further contribute to efficacy and toxicity when rifabutin is given together with a PI; however, this relation has not yet been adequately evaluated.

The method of using a pooled analysis was cumbersome but successful. Pooled population analysis is an established statistical technique closely associated with meta-analysis and systematic reviews of the literature. The concepts of statistical meta-analysis are increasingly applied to pharmacometric analysis (model-based meta-analysis).^{28,72,73} This method enables the evaluation of new or additional questions compared with the original studies and it is cost-effective because no new subjects are recruited. The coalescing of data at the individual subject level may increase precision of pharmacokinetic parameters and provide more power for effect and/or covariate detection.²⁸ Limitations of this approach are similar to limitations of traditional meta-analysis, including the analysis of only selected available studies ('file drawer problem'), and also include reliance on source data of individual subjects in published studies. The present method is time-consuming and may introduce errors because of increased variability observed during the analysis. Furthermore, addition of information from multiple studies may increase heterogeneity. We considered adding inter-study variability, but this was not feasible because of the prohibitive runtimes of the model. Similarly, estimation of within-subject variability was considered, but was not feasible.

Limitations of the present study include insufficient data to describe all subpopulations sufficiently, especially because of the model's complexity, multiple effects of drug–drug interactions and large variability. We were unable to detect pharmacokinetic differences between patients who had TB alone versus TB/HIV; despite the large population, there were few patients who had TB only, and we could not differentiate between the effects of different PIs. Furthermore, the data were sparse for many concomitant PIs, making it necessary to broadly categorize these drugs into two groups (non-boosted and ritonavir-boosted PIs). Each PI may have different interaction potential, especially non-boosted PIs. However, this limitation may be less relevant because PIs boosted with 100 mg of ritonavir likely have an equivalent interaction potential to each other, and non-boosted PIs are infrequently used in areas where TB is endemic.

Further research is important because rifabutin is now on the WHO Essential Medicines List⁷⁴ for TB therapy in HIV-infected patients. Rifabutin is also important for treatment and prevention of atypical mycobacteria, especially *Mycobacterium avium*, in HIV-coinfected patients. The safety of regimens that combine PIs and rifamycins must be evaluated in comparison with other available therapeutic options.

In conclusion, we showed that drug–drug interaction between rifabutin and PIs may cause increased exposure to rifabutin by 210% for TB/HIV patients with non-boosted PIs and 280% with ritonavir-boosted PIs; for healthy volunteers, the increase was >300% and >400%. The dosing regimens that result in similar exposure (C_{av_ss} , C_{max_ss}) to 300 mg of rifabutin given alone are 150 mg of rifabutin given once daily with a non-boosted PI or 300 mg of rifabutin every 3 days when given together with a ritonavir-boosted PI. Predicted peak concentrations suggested that with dosages of 150 mg of rifabutin daily with ritonavir-boosted PIs, the average patient is unlikely to experience rifabutin exposure >1 mg/L, which is a concentration limit that has been associated with toxicity. Therefore, daily instead of intermittent dosing with 150 mg of rifabutin in TB/HIV-coinfected patients

may be appropriate in patients who are monitored adequately for toxicity. New formulations or strengths of rifabutin may be required for more optimized dosing regimens.

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Author contributions

S. H. collected data, performed the analysis, interpreted results and drafted the manuscript. E. M. S. and R. N. contributed to data analysis and manuscript review. H. M. and M. O. K. developed the study concept, interpreted study results and reviewed the manuscript. B. F., M. H. W., S. B., C. A. P., K. G., C. F., A. P., P. V. and P. L. O. contributed data and reviewed the manuscript.

Disclaimer

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Supplementary data

Supplementary data, including Tables S1 to S4 and Figure S1, are available at JAC Online (<http://jac.oxfordjournals.org/>).

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